

Grouping APEC Countries

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1 RSR

- Methodology
- Result

2 *K*-Means

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- Result

- RSR is a statistical method used for multi-criteria decision-making (MCDM). It was proposed by Professor Tian Fengtiao in 1988. The method combines the ranks of multiple indicators into a single composite score (RSR), which is then used to evaluate and rank or group alternatives. RSR is particularly useful in fields like healthcare, statistical quality control, and environmental management.

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Key Concepts

Rank The rank of a sample.

Sum The sum of ranks across multiple criteria.

Ratio The ratio of the sum of ranks to the total possible ranks, resulting in a dimensionless statistic.

Step 1: Ranking

In this step we rank each indicator for each object. Suppose we have m objects and n indicators.

For positive indicators: $R_{ij} = 1 + (m - 1) \cdot \frac{X_{ij} - X_{j\min}}{X_{j\max} - X_{j\min}}.$

For negative indicators: $R_{ij} = 1 + (m - 1) \cdot \frac{X_{j\max} - X_{ij}}{X_{j\max} - X_{j\min}}.$

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For negative indicators: $R_{ij} = 1 + (m - 1) \cdot \frac{X_{j\max} - X_{ij}}{X_{j\max} - X_{j\min}}$.

Step 2: Calculate RSR

For the i -th object: $RSR_i = \frac{\sum_{j=1}^n R_{ij}}{m \cdot n}$.

If weights are assigned to indicators, calculate the Weighted RSR

(WRSR): $WRSR_i = \frac{\sum_{j=1}^n w_j \cdot R_{ij}}{m \cdot n}$.

Step 3: Determine RSR Distribution and Calculate Probit

Here we create a frequency distribution table (Table 1) for RSR values and then calculate cumulative frequencies and convert them to probability units (*Probit*) using the standard normal distribution.

$$Probit = \text{norminv}(p, 0, 1) + 5$$

where

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$$\mu := \text{norminv}(p, 0, 1)$$

Step 4: Linear Regression

Perform a linear regression analysis with *Probit* as the independent variable and *RSR* (or *WRSR*) as the dependent variable.

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Step 5: Grouping and Ranking

First, *RSR* values for different *Probit* levels are estimated using the regression equation. Then objects are grouped into categories based on their estimated *RSR* values. In our specific case, APEC countries are divided into 3 groups. The grouping is often done using standard normal distribution percentiles (e.g., 15.87%, 50%, 84.13%).

After grouping, variance of *RSR* in each group is calculated to help validate whether the grouping is appropriate.

Grouping Results

RSRw	f	Σf	R	$\bar{R}$	$\frac{\bar{R}}{m} \times 100\%$	Probit	Estimated RSRw	Group	variance	Countries
0.725691	1	1	1	1	5.263158	3.380144	0.553475	I	0.012629	MY
0.493747	1	2	2	2	10.526316	3.747880	0.500337	I	0.012629	JP
0.481333	1	3	3	3	15.789474	3.996852	0.464360	I	0.012629	AU
0.405734	1	4	4	4	21.052632	4.195404	0.435669	II	0.006668	NZ
0.381532	1	5	5	5	26.315789	4.366360	0.410966	II	0.006668	CA
0.364845	1	6	6	6	31.578947	4.520494	0.388694	II	0.006668	KR
0.351720	1	7	7	7	36.842105	4.663962	0.367962	II	0.006668	US
0.337927	1	8	8	8	42.105263	4.800799	0.348189	II	0.006668	SG
0.330010	1	9	9	9	47.368421	4.933988	0.328943	II	0.006668	BN
0.296886	1	10	10	10	52.631579	5.066012	0.309866	II	0.006668	RU
0.268075	1	11	11	11	57.894737	5.199201	0.290620	II	0.006668	CN
0.190594	1	12	12	12	63.157895	5.336038	0.270847	II	0.006668	CL
0.187926	1	13	13	13	68.421053	5.479506	0.250116	II	0.006668	PH
0.180838	1	14	14	14	73.684211	5.633640	0.227843	II	0.006668	MX
0.179588	1	15	15	15	78.947368	5.804596	0.203140	II	0.006668	ID
0.179346	1	16	16	16	84.210526	6.003148	0.174449	III	0.000837	TH
0.161907	1	17	17	17	89.473684	6.252120	0.138472	III	0.000837	VN
0.122133	1	18	18	18	94.736842	6.619856	0.085334	III	0.000837	PE
0.107841	1	19	19	19	98.684211	7.221520	-0.001607	III	0.000837	PG

Table: 2019stats

Grouping Results

Group	Countries
I	MY JP AU
II	NZ CA KR US SG BN RU CN CL PH MX ID
III	TH VN PE PG

Table: 3 groups of 2019

Grouping Results

RSRw	f	Σf	R	$\bar{R}$	$\frac{\bar{R}}{m} \times 100\%$	Probit	Estimated RSRw	Group	variance	Countries
0.687744	1	1	1	1	5.263158	3.380144	0.688367	I	0.000605	JP
0.672290	1	2	2	2	10.526316	3.747880	0.625427	I	0.000605	AU
0.629583	1	3	3	3	15.789474	3.996852	0.582815	I	0.000605	NZ
0.538412	1	4	4	4	21.052632	4.195404	0.548832	II	0.011605	CA
0.529926	1	5	5	5	26.315789	4.366360	0.519572	II	0.011605	KR
0.494267	1	6	6	6	31.578947	4.520494	0.493191	II	0.011605	US
0.457257	1	7	7	7	36.842105	4.663962	0.468636	II	0.011605	PH
0.450211	1	8	8	8	42.105263	4.800799	0.445216	II	0.011605	RU
0.440038	1	9	9	9	47.368421	4.933988	0.422420	II	0.011605	SG
0.420038	1	10	10	10	52.631579	5.066012	0.399824	II	0.011605	BN
0.363009	1	11	11	11	57.894737	5.199201	0.377028	II	0.011605	CN
0.355538	1	12	12	12	63.157895	5.336038	0.353607	II	0.011605	MY
0.243755	1	13	13	13	68.421053	5.479506	0.329052	II	0.011605	MX
0.242805	1	14	14	14	73.684211	5.633640	0.302672	II	0.011605	CL
0.214867	1	15	15	15	78.947368	5.804596	0.273412	II	0.011605	TH
0.204034	1	16	16	16	84.210526	6.003148	0.239429	III	0.000882	ID
0.185700	1	17	17	17	89.473684	6.252120	0.196816	III	0.000882	VN
0.177442	1	18	18	18	94.736842	6.619856	0.133877	III	0.000882	PE
0.124176	1	19	19	19	98.684211	7.221520	0.030899	III	0.000882	PG

Table: 2015stats

Grouping Results

Group	Countries
I	JP AU NZ
II	CA KR US PH RU SG BN CN MY MX CL TH
III	ID VN PE PG

Table: 3 groups of 2015

Grouping Results

RSRw	f	Σf	R	$\bar{R}$	$\frac{\bar{R}}{m} \times 100\%$	Probit	Estimated RSRw	Group	variance	Countries
0.710265	1	1	1	1	5.263158	3.380144	0.695696	I	0.001440	AU
0.689659	1	2	2	2	10.526316	3.747880	0.641049	I	0.001440	JP
0.621467	1	3	3	3	15.789474	3.996852	0.604050	I	0.001440	NZ
0.574576	1	4	4	4	21.052632	4.195404	0.574544	II	0.006312	US
0.559670	1	5	5	5	26.315789	4.366360	0.549139	II	0.006312	CA
0.532829	1	6	6	6	31.578947	4.520494	0.526234	II	0.006312	BN
0.497873	1	7	7	7	36.842105	4.663962	0.504914	II	0.006312	PH
0.476371	1	8	8	8	42.105263	4.800799	0.484580	II	0.006312	SG
0.444123	1	9	9	9	47.368421	4.933988	0.464787	II	0.006312	KR
0.398566	1	10	10	10	52.631579	5.066012	0.445168	II	0.006312	RU
0.391493	1	11	11	11	57.894737	5.199201	0.425375	II	0.006312	MY
0.390604	1	12	12	12	63.157895	5.336038	0.405041	II	0.006312	MX
0.388318	1	13	13	13	68.421053	5.479506	0.383721	II	0.006312	CL
0.373806	1	14	14	14	73.684211	5.633640	0.360815	II	0.006312	CN
0.310665	1	15	15	15	78.947368	5.804596	0.335410	II	0.006312	VN
0.269977	1	16	16	16	84.210526	6.003148	0.305905	III	0.000947	TH
0.255272	1	17	17	17	89.473684	6.252120	0.268906	III	0.000947	PE
0.240514	1	18	18	18	94.736842	6.619856	0.214259	III	0.000947	ID
0.188394	1	19	19	19	98.684211	7.221520	0.124849	III	0.000947	PG

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Grouping Results

Group	Countries
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Table: 3 groups of 2013

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RSRw	f	Σf	R	$\bar{R}$	$\frac{\bar{R}}{m} \times 100\%$	Probit	Estimated RSRw	Group	variance	Countries
0.589870	1	1	1	1	5.263158	3.380144	0.656474	I	0.000181	JP
0.583047	1	2	2	2	10.526316	3.747880	0.604152	I	0.000181	AU
0.558511	1	3	3	3	15.789474	3.996852	0.568728	I	0.000181	PH
0.545549	1	4	4	4	21.052632	4.195404	0.540478	II	0.005907	US
0.542085	1	5	5	5	26.315789	4.366360	0.516154	II	0.005907	KR
0.495233	1	6	6	6	31.578947	4.520494	0.494223	II	0.005907	CA
0.480501	1	7	7	7	36.842105	4.663962	0.473810	II	0.005907	RU
0.479373	1	8	8	8	42.105263	4.800799	0.454341	II	0.005907	SG
0.475111	1	9	9	9	47.368421	4.933988	0.435391	II	0.005907	MY
0.448377	1	10	10	10	52.631579	5.066012	0.416606	II	0.005907	NZ
0.432454	1	11	11	11	57.894737	5.199201	0.397656	II	0.005907	VN
0.422379	1	12	12	12	63.157895	5.336038	0.378186	II	0.005907	CN
0.385966	1	13	13	13	68.421053	5.479506	0.357773	II	0.005907	MX
0.342735	1	14	14	14	73.684211	5.633640	0.335843	II	0.005907	BN
0.269255	1	15	15	15	78.947368	5.804596	0.311519	II	0.005907	ID
0.256418	1	16	16	16	84.210526	6.003148	0.283269	III	0.002164	CL
0.172564	1	17	17	17	89.473684	6.252120	0.247844	III	0.002164	PE
0.170204	1	18	18	18	94.736842	6.619856	0.195522	III	0.002164	TH
0.128254	1	19	19	19	98.684211	7.221520	0.109917	III	0.002164	PG

Table: 2009stats

Grouping Results

Group	Countries
I	JP AU PH
II	US KR CA RU SG MY NZ VN CN MX BN ID
III	CL PE TH PG

Table: 3 groups of 2009

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- The algorithm is computationally efficient and can handle large datasets, making it suitable for applications like similarity grouping, nonlinear prediction, and approximating multivariate distributions.

Problem Formulation

- Given: a dataset $\{z_1, z_2, \dots, z_n\}$ where each z_i is a point in N -dimensional space ($z_i \in \mathbb{R}^N$) and an integer k (the number of clusters).

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- The objective is to find a partition $S = \{S_1, S_2, \dots, S_k\}$ of the dataset such that the within-cluster variance is minimized. The within-cluster variance is defined as:

$$W(S) = \sum_{i=1}^k \sum_{z \in S_i} \|z - \mu_i\|^2$$

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- μ_i is the centroid of cluster S_i , defined as $\mu_i = \frac{1}{|S_i|} \sum_{z \in S_i} z$

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- where:
- μ_i is the centroid of cluster S_i , defined as $\mu_i = \frac{1}{|S_i|} \sum_{z \in S_i} z$,
- $\|z - \mu_i\|^2$ is the squared Euclidean distance between point z and centroid μ_i .

Basic Steps

- The K -means algorithm iteratively refines the clusters by alternating between two steps, which are repeated until the centroids no longer change significantly, or a maximum number of iterations is reached.

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Step 1: Assignment

$$S_i = \{z : |z - \mu_i| \leq |z - \mu_j| \quad \forall j \neq i\}$$

This creates a minimum distance partition of the data, where each data point z is assigned to the cluster S_i whose centroid is closest.

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This creates a minimum distance partition of the data, where each data point z is assigned to the cluster S_i whose centroid is closest.

Step 2: Update

Recompute the centroids μ_i as the mean of all points assigned to cluster S_i :

$$\mu_i = \frac{1}{|S_i|} \sum_{z \in S_i} z$$

Convergence

The sequence of within-cluster variances $W(x^n)$ converges almost surely to a limit W_∞ , which corresponds to the variance of an optimal partition.

Formally:

$$W_\infty = \lim_{n \rightarrow \infty} W(x^n)$$

where x^n represents the centroids at iteration n .

Theoretical Foundations

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Formally:

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Unbiased Centroids

A k -tuple of centroids $x = (x_1, x_2, \dots, x_k)$ is called unbiased if each centroid x_i is the true mean of its cluster S_i : $x_i = \mu_i$.

The K -means algorithm converges to a set of unbiased centroids under certain conditions.

Grouping Results

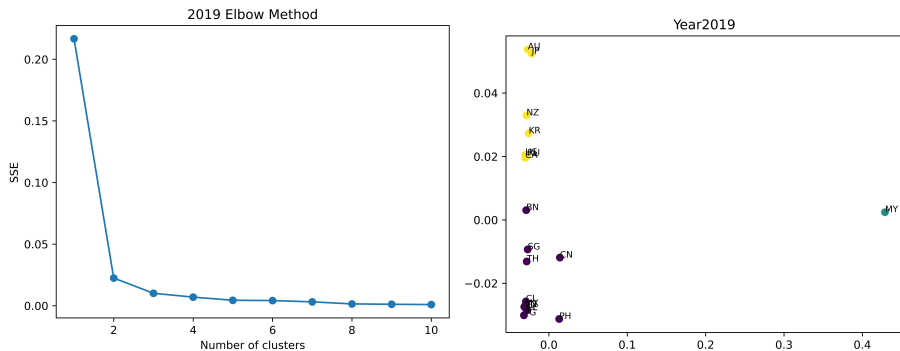


Figure: 2019 clustering

Grouping Results

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I	AU JP MY NZ KR RU
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Table: 2019 clusters

Grouping Results

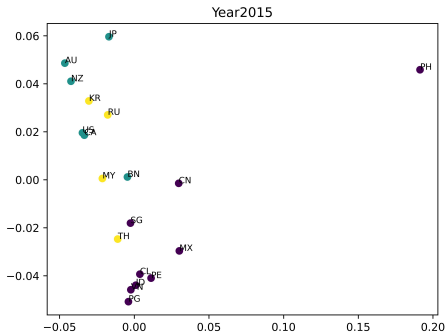
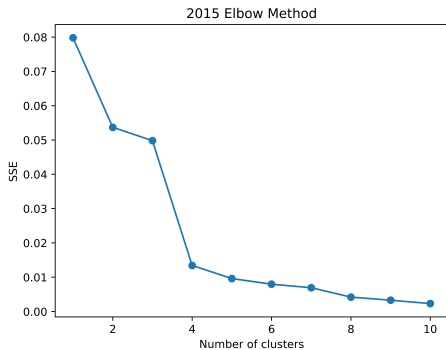


Figure: 2015 clustering

Grouping Results

Group	Countries
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Grouping Results

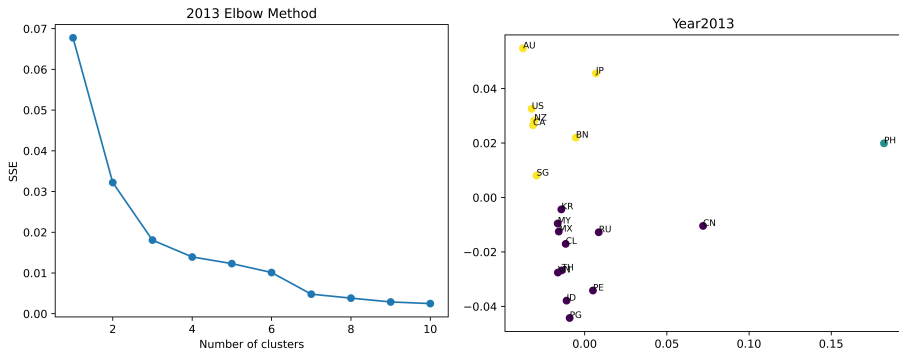


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Table: 2013 clusters

Grouping Results

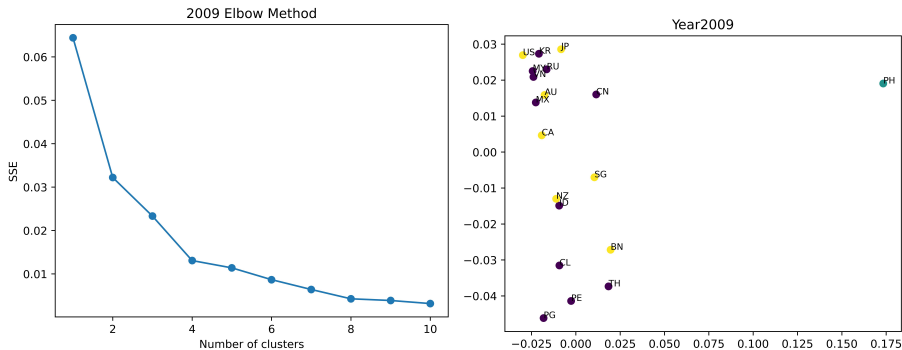


Figure: 2009 clustering

Grouping Results

Group	Countries
I	ID JP MY MX CN KR RU US VN
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Reflection on K -Means

- The grouping result of this method differs significantly from that of the RSR method.

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- In K -Means, we are actually supposed to use "elbow method" to determine the number of clusters.
- The dataset is not large enough for machine learning algorithms such as K -Means to produce satisfying results.

The End